A Novel Sensorless Hybrid MPPT Method Based on FOCV Measurement and P&O MPPT Technique for Solar PV Applications

Mithun Krishnan M
Department of Electrical & Electronics Engineering
Amrita Vishwa Vidyapeetham
Amritapuri, India
Email id: mkmithun1995@gmail.com

Bharath K R
Department of Electrical & Electronics Engineering
Amrita Vishwa Vidyapeetham
Amritapuri, India
Email id: bharathkr@am.amrita.edu

***Abstract*—In the era of Renewable Energy, Solar power is considered as one among the prominent and emerging energy source. But the output of a solar PV varies drastically due to varying atmospheric conditions like temperature and solar insolation. The Maximum Power Point Tracking (MPPT) method is an inevitable part which must be present for the efficient utilization of solar energy tapped using solar PV units. The conventional methods fail to trail the maximum available power efficiently and economically, and can even fail during partial shading conditions. This paper proposes an advanced hybrid algorithm that blends the advantages of both Fractional Open-Circuit Voltage (FOCV) MPPT and Perturb and Observe (P&O) algorithm, thereby overcoming their inherited flaws. To validate the proposed algorithm's potency, the technique is simulated using Simulink environment on MATLAB and simulation results show a clear improvement in achieving the MPPT under varying atmospheric conditions.**

***Keywords*—Solar PV, P&O MPPT, FOCV MPPT, Hybrid MPPT, Buck Converter**

I. Introduction

In this fast-growing world, electricity demand is rapidly thriving day-by-day. The limited availability of conventional resources and their harmful effects made the human kind think about an unconventional and sustainable source of energy. Renewable Energy Sources (RES) are the future for powering the whole nation. Solar, Wind, Biomass, Geothermal are a few among the list. Among the renewables, energy from the sun is very promising. The main features of such a promising energy source are easy availability, eco-friendly nature, less operational and maintenance cost.

The performance of PV panel depends upon the ambient cell temperature and solar insolation. Any variation in the above-mentioned parameters disturbs the desired performance of the entity. The drawbacks of PV panels are low conversion efficiency and non-linearity in I-V characteristics. The Maximum Power Point Tracking (MPPT) [1] method is adopted so as to accurately track the maximum available power from a solar panel. The primary objective of MPPT controller is to achieve maximum available power from the photovoltaic panels irrespective of the atmospheric parameters like solar irradiance and temperature. Such controllers grant the desired amount of power, thereby curtailing the cost and enhancing the system reliability. Hill Climbing method, Perturb and Observe method (P&O) [2], Artificial Neural Network (ANN) [3], Fractional Open Circuit Voltage algorithm (FOCV) [4], Incremental Conductance method (Inc- Cond) [5] [6] and Particle Swarm Optimization (PSO) [7] are the widely used MPPT methods. Though these controllers serve the same objective, these algorithms vary in terms of PV array dependency, control strategies, number of control variables, implementation complexity, application, cost, oscillations around the MPP point, speed and types of circuitry. MPPT is considered to be one of the most efficient mechanisms to enhance the performance of a solar panel. These conventional MPPT algorithms may even fail during Partial Shading Conditions (PSC). This necessitates the demand for advanced MPPT controllers [8]. To further improve the effectiveness of these conventional methods, [9] proposes an improved adaptive P&O MPPT method, which largely minimizes the steady-state oscillation, inaccurate tracking direction and instills the ability to detect the global power point during PSC. FOCV based MPPT is another method that makes use of open circuit voltage of solar PV. This method needs only a single sensor and requires only minor computations when compared with other conventional MPPT methods, but fails during subtle changes in temperature and irradiance level. The shading of uneven pattern results in numerous spikes in the P-V characteristics. This problem due to shading effects is mitigated and discussed in [10]. [11] discusses a novel MPPT which combines the relation between the load curve and the IV characteristics of a solar module and has a rapid convergence speed.

This work is coordinated into six sections. Section II discusses the one diode model of a solar PV system and its characteristics under varying atmospheric conditions. Section III introduces conventional MPPT methods like P&O and FOCV techniques are discussed, along with the proposed hybrid MPPT techniques. In section IV implementation of the complete system in MATLAB/Simulink environment is briefed. Section V details simulation results and analyses the results by comparing and contrasting with the existing conventional methods. The Paper is concluded in section VI. The future expansion possibilities of the proposed system are also discussed in the concluding segment of this paper.

978-1-7281-3250-1/19/$31.00 © 2019 IEEE

Authorized licensed use limited to: Universite de Technologie de Compiegne (UTC). Downloaded on April 08,2024 at 16:43:14 UTC from IEEE Xplore. Restrictions apply.

II. Solar Photo Voltaic System

A. One Diode Model

The photovoltaic effect is the basic principle that makes a PV cell convert incident solar radiation (sunlight) into electricity. Solar radiation, which constitutes infrared heat waves and visible light waves, when getting incident on a photovoltaic cell get absorbed, reflected, or even go right through [12]. During this phenomenon, the light that is absorbed develops useful electricity. Each category of the photovoltaic panel has its own distinct attributes correlating to the atmospheric parameters like temperature and solar irradiance. These cells are either connected in parallel, series or series-parallel array combinations that might depend upon the end-user application. If they are connected in series, the net voltage gets added up and in a parallel combination, the overall current delivered is summed up based on the number of cells or PV array connected, and while in series-parallel combination, required voltage and current levels can be achieved.

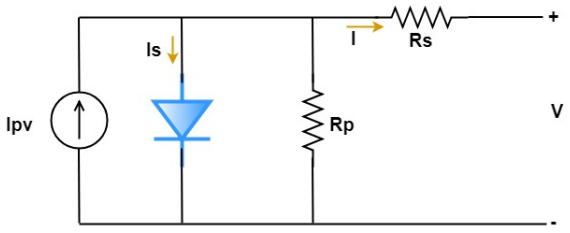


Fig. 1. One Diode model of Solar Photo Voltaic system

The solar cell is modeled based on the IV characteristics of a solar PV panel, given in equation 1 [13].

$$I = I_{pv} - I_s \left(exp\left(\frac{q(V + IR_s)}{N_s kTa} \right) - 1 \right) - \frac{V + R_s I}{R_p} \quad (1)$$

Where:

I_{pv}	: Photo Voltaic Current (A)
I_s	: Reverse Saturation Current (A)
q	: Charge of Electron ($1.602176 \times 10^{-19} C$)
k	: Boltzmann's constant ($1.3806 \times 10^{-23} J/K$)
a	: Ideality constant for Diode
R_s	: Resistance in Series with the cell (Ω)
R_p	: Resistance in Parallel with the cell (Ω)
N_s	: Number of Cells in series
T	: Atmospheric Temperature (K)

The equivalent one diode model of the solar PV cell is given in fig. 1.

III. Conventional MPP Tracking Methods

The highest efficiency ever recorded for a commercially available solar PV on the current market is around 22.5%, while most of the panels possess an efficiency range from 15% to 17%. The highest efficiency of 44.4% was claimed by researchers [14], by using a concentrator triple-junction compound solar cell. To trap the maximum power from a non-linear power source, MPPT methods are widely accepted. To extract the maximum available power from a solar panel, MPPT makes use of different schemes in switching the power electronic converter. The following techniques are some of the extensively accepted and commonly used MPPT methods [15] implemented for various solar PV applications [16].

A. Perturb & Observe (P&O) MPPT

Perturb & Observe [14] MPPT technique is one of the extensively adopted iterative MPPT methods to trap maximum power. This method requires two sensors, namely the current, and a voltage sensor, that periodically measures the current and voltage of a SPV cell to calculate the system power. It starts by perturbing along the operating values and observes any variation in the power difference. If the variation is positive, then the step size is moved in the same direction and vice-versa.

This method is known for its simplicity and economical. Even, the previous knowledge of the solar PV is not required. The technique fails during an abrupt change in atmospheric conditions [17] like the sudden variation in irradiance level or partial shading.

B. Fractional Open Circuit Voltage (FOCV) MPPT

At a given atmospheric condition, there exists an exclusive operating point P(Vmp, Imp) called the Maximum Power Point (MPP) at which the solar PV power is at its peak(Pmp). If anyone of the point (Vmp or Imp) can be tracked by some methods, then corresponding Pmp can be found out at ease. In the FOCV measurement, the nonlinear IV curve of a solar PV is modeled using mathematical equations or basic approximations under varying atmospheric conditions. Based on this understanding, an empirical relationship is calculated as follows:

$$Vmp \approx K \times Voc \quad (2)$$

It is found that the K value lies between 0.71 to 0.78 times the Voc value [10] [18]. In this technique, the solar panel is open-circuited and Voc is monitored for a fraction of second.

C. Proposed MPPT Technique

The proposed MPPT algorithm is designed with more emphasis on the conventional P&O MPPT while taking aid from FOCV method [19] during crucial and chaotic conditions. The solar panel voltage is audited and stored when the solar panel is open-circuited. Based on some experimental results, it was found that the voltage at MPP of a PV falls between (0.71-0.78) times the open circuit voltage of the solar PV [10]. An added advantage of the proposed algorithm is that it combines the benefits of both FOCV and P&O MPPT [20] while overcoming their inherited deficiencies. The proposed MPPT method works in the following modes:

- Mode 1: Estimating the MPP using Voc;
- Mode 2: Variable Step P&O loop mode for MPPT.

Mode1: Estimating the MPP using Voc

The advanced hybrid MPPT algorithm launches by the calculation of MPP using Voc. As the loop progress, energy abstracted from the solar PV is interrupted for a short duration by yielding sufficient switching sequences to the power-electronic converter. Once the PV module becomes open-circuited, the panel voltage is measured and saved. The value of Vmp is calculated from equation no. 2 and the error difference between Voc and Vmp are calculated. The outcome

Authorized licensed use limited to: Universite de Technologie de Compiegne (UTC). Downloaded on April 08,2024 at 16:43:14 UTC from IEEE Xplore. Restrictions apply.

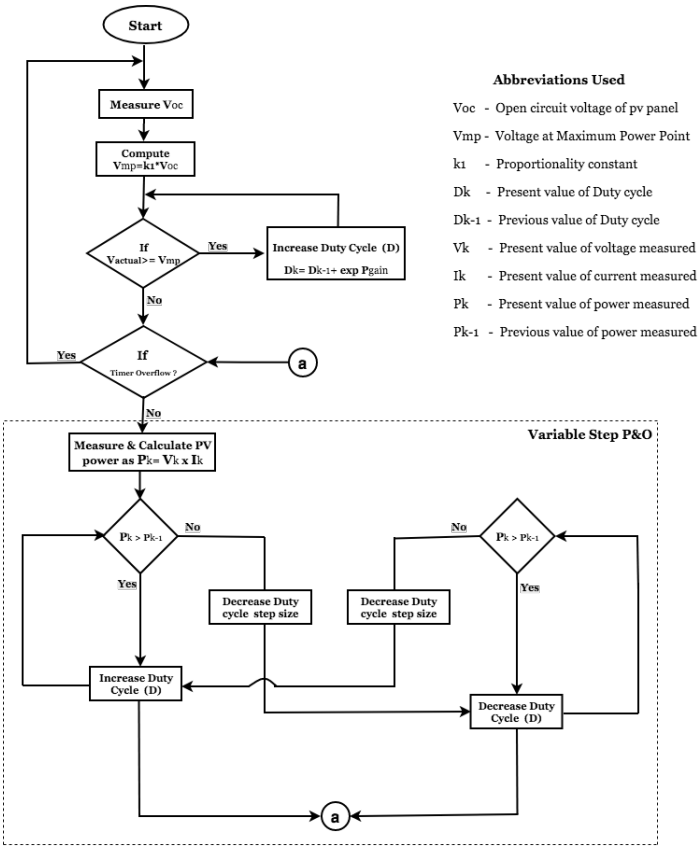


Fig. 2. Flowchart of proposed hybrid MPPT algorithm

is then fed to the compensator to determine the duty cycle. This entire process completes with minimal or no effect on the load which is fed from the solar PV system.

There can be subtle changes in temperature and solar irradiation level as the time passes. For a solar panel, Vmp is dependent on Voc, irradiance level, and atmospheric temperature. To compensate with this rapidly varying parameters and to improve the effectiveness of FOCV method, a timer overflow concept in the microcontroller is implemented. During the course of an interrupt which is a regular timely event, Voc of the PV system is found out and stored. Depending upon the updated Voc value which gets periodically updated, Vmp value is estimated from the information fetched from the infinite loop. This helps in proper compensation accompany the change in Voc due to the subtle changes in atmospheric conditions. Whenever the error value is greater than or equal to zero, the duty pulses and power values are saved and the system reaches the P&O loop.

Mode 2: Variable Step P&O loop mode for MPPT

At this stage, the net power of the solar PV is computed based on the current and voltage levels measured by the current and voltage sensors respectively. Then, the computed power is compared with the previous power. The panel operates in the same direction (decreasing or increasing PV current/voltage by varying duty cycle) if the calculated power is higher. The operation shifts to the next loop and direction of operation are reversed once a flip occurs. The system

TABLE I. SPECIFICATION OF THE PV ARRAY CONSIDERED FOR RESEARCH

Description	Rating
Maximum Power (Pmp)	100 W
Maximum Current (Imp)	5 A
Maximum Voltage (Vmp)	23 V
Open-Circuit Voltage (Voc)	28.5 V
Short-Circuit Current (Isc)	5.49 A
Number of parallely connected modules	1
Number of series strings	1

easily reaches the MPP and continues to float between the loop. To mitigate this issue, the step size is adjusted once the panel reaches MPP. After a particular time, the timer value resets and operation is transferred to FOCV to accumulate the change in atmospheric conditions.

IV. SIMULATION ARRANGEMENTS

For the analysis, the overall system is modeled and simulated in the MATLAB/Simulink environment. Fig. 3 shows the proposed PV system [6] employed of PV array, Buck converter with the proposed MPPT algorithm and a constant DC Load. A Solar photovoltaic system with the specifications mentioned as in Table I is considered for the simulation purpose in MATLAB/Simulink.

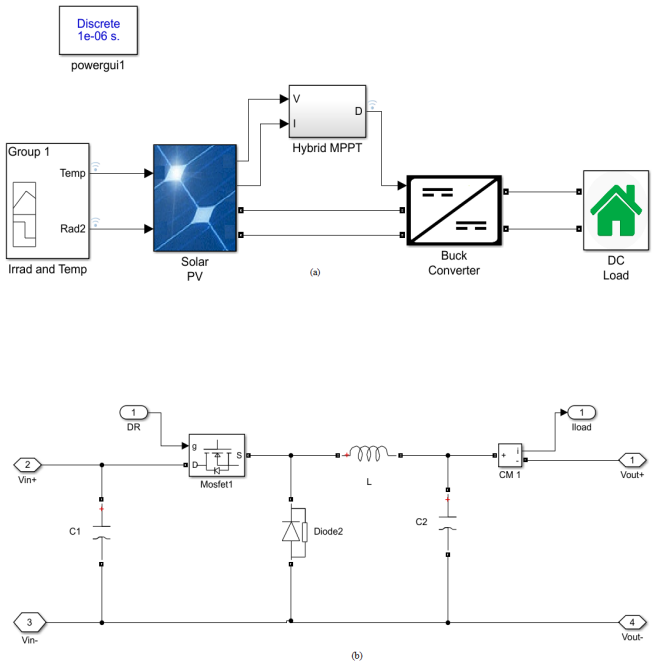


Fig. 3. MATLAB/Simulink model of (a) Complete System, (b) Buck converter

The solar panel characteristics are depicted in fig. 4. The output of the SPV module is fed as an input to the step-down converter, which reduces the output voltage by altering the duty cycle provided by the proposed hybrid controller. To validate the performance of the hybrid algorithm, the same system was simulated for conventional MPPTs like P&O and FOCV and the results are compared. The specifications of the buck converter used for the analysis is shown in Table II.

Authorized licensed use limited to: Universite de Technologie de Compiegne (UTC). Downloaded on April 08,2024 at 16:43:14 UTC from IEEE Xplore. Restrictions apply.

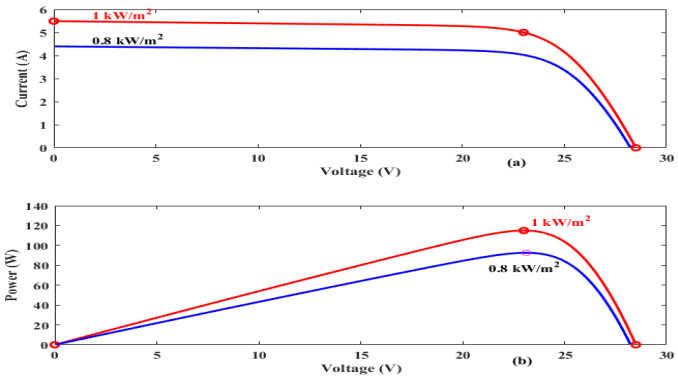


Fig. 4. Characteristics of PV module considered for research (a) I-V Curve and (b) P-V curve

TABLE II. BUCK CONVERTER SPECIFICATIONS

Description	Rating
PV panel rating	23 V, 5 A, 100 W
Inductor	L1= 17.66 uH
DC link Capacitor	C1= 100 uF
Load Capacitor	C2= 500 uF
Load Resistance	1 ohm
Switching frequency	20 kHz

V. SIMULATION RESULTS

To validate the potentiality of the proposed algorithm, the solar irradiance was varied between 1000 to 800 W/m^2 with a constant temperature of 25°C. The buck converter steps down the input voltage from 23V to 12V under normal irradiance. The output voltage drastically reduced when the irradiance value was reduced to 800 W/m^2. The proposed hybrid MPPT method could track the MPP successfully even during varying irradiance. The output load characteristics of the buck converter is provided in fig. 5.

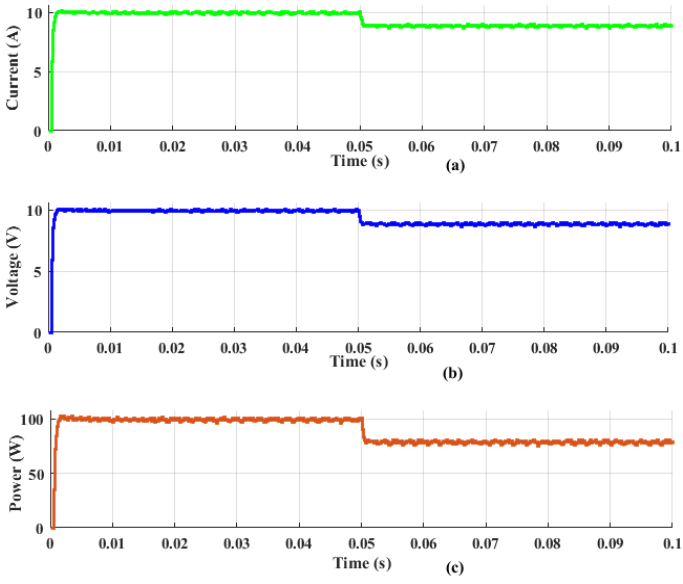


Fig. 5. Plot for Load Characteristics of DC Load, (a) Current vs time (b) Voltage vs time and (c) Power vs time

To compare the performance of the hybrid MPPT algorithm, the same model was simulated using P&O and FOCV based MPPT. Fig. 6 shows the comparison between the three MPPT algorithms. The solar irradiance level was shifted from 1000 to 800 W/m^2 at 0.01s. The simulation results show that the hybrid MPPT method exceeds the conventional algorithms in terms of ripple content, tracking time and efficiency of the system. A detailed analysis results are given in Table III.

TABLE III. COMPARISON OF P&O, FOCV AND PROPOSED HYBRID MPPT

	P&O MPPT	FOCV MPPT	Hybrid MPPT
Tracking Time (ms)	0.32	0.21	0.16
Ripple Content (%)	14	4	2
Efficiency (%)	85.4 (1000 W/m^2) 72.78 (800 W/m^2)	87.8 (1000 W/m^2) 73.82 (800 W/m^2)	98.17 (1000 W/m^2) 81.67 (800 W/m^2)

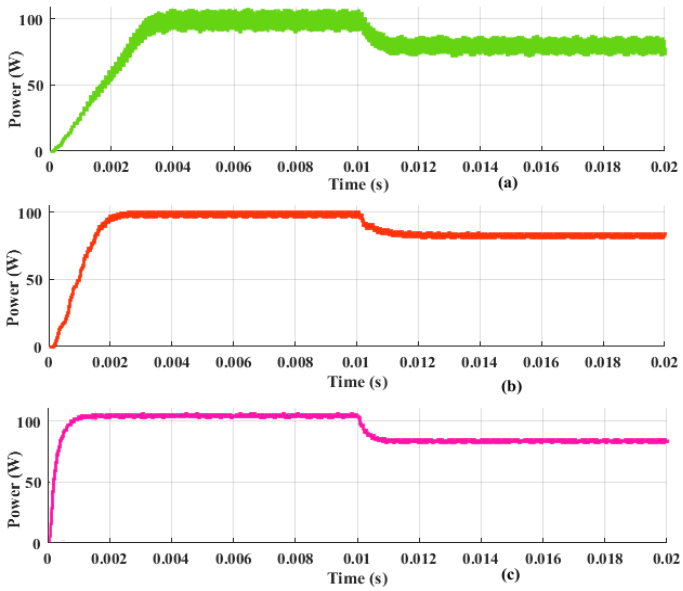


Fig. 6. Plot for power vs time for various MPPT techniques (a) P&O MPPT (b) FOCV Method (c) Hybrid MPPT

VI. CONCLUSION

The RES is advantageous than any other energy resources, as it is suited for any individual application or for the grid. The effective utilization of solar energy and the MPPT controller is proposed in this paper. For the analysis, a Buck converter with hybrid MPPT has been successfully simulated. The open-circuit voltage of a solar PV is measured by the intelligent decision making capability of the advanced hybrid MPPT method.

The Hybrid MPPT method dominates over the traditional MPPT algorithms by the following factors:

- Implementation is more effortless.
- It has a fair convergence speed under rapidly varying atmospheric conditions.
- It reaches the MPP region rapidly.

Authorized licensed use limited to: Universite de Technologie de Compiegne (UTC). Downloaded on April 08,2024 at 16:43:14 UTC from IEEE Xplore. Restrictions apply.

- It achieves more efficiency and fewer oscillations in power without reducing the tracking speed of P&O, by utilizing reduced perturbation size.
- The Voc value is intelligently sensed during varying atmospheric conditions, without implementing an additional sensor to measure the irradiance level.
- The proposed technique is much more efficient than the conventional methods to harness energy.

During the initial stage, to make the system more close to the peak power point, an offline MPPT method is utilized, and then using a variable step-size P&O MPPT technique, the MPP is reached more efficiently and economically. The proposed hybrid algorithm hybridizes the above-mentioned techniques to track MPP more accurately and can even be utilized for a wide range of solar PV applications. The PV power tracking research area is endowed with the proposed hybrid algorithm.

REFERENCES

[1] B. Subudhi and R. Pradhan, "A comparative study on maximum power point tracking techniques for photovoltaic power systems," *IEEE Transactions on Sustainable Energy*, vol. 4, pp. 89–98, Jan 2013.

[2] W. Zhu, L. Shang, P. Li, and H. Guo, "Modified hill climbing mppt algorithm with reduced steady-state oscillation and improved tracking efficiency," *The Journal of Engineering*, vol. 2018, no. 17, pp. 1878–1883, 2018.

[3] H. Choutapalli, K. R. Bharath, and P. Kanakasabapathy, "A review on advanced mppt methods for spv system under partial shaded conditions," in *2018 International Conference on Control, Power, Communication and Computing Technologies (ICCPCCT)*, pp. 305–310, March 2018.

[4] D. C. Huynh and M. W. Dunnigan, "Development and comparison of an improved incremental conductance algorithm for tracking the mpp of a solar pv panel," *IEEE Transactions on Sustainable Energy*, vol. 7, pp. 1421–1429, Oct. 2016.

[5] K. P. K. Vineeth and C. A. Asha, "An efficient solar power converter with high mpp tracking accuracy for rural electrification," in *2014 International Conference on Computation of Power, Energy, Information and Communication (ICCPEIC)*, pp. 383–389, April 2014.

[6] K. P. K. Vineeth and M. K. Sreenivasan, "Hardware implementation of high efficient and high voltage gain dc-dc converter for dc microgrid applications," in *2015 IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT)*, pp. 1–5, March 2015.

[7] R. B. A. Koad, A. F. Zobaa, and A. El-Shahat, "A novel mppt algorithm based on particle swarm optimization for photovoltaic systems," *IEEE Transactions on Sustainable Energy*, vol. 8, pp. 468–476, April 2017.

[8] H. M. El-Helw, A. Magdy, and M. I. Marei, "A hybrid maximum power point tracking technique for partially shaded photovoltaic arrays," *IEEE Access*, vol. 5, pp. 11900–11908, 2017.

[9] J. Ahmed and Z. Salam, "An enhanced adaptive p&o mppt for fast and efficient tracking under varying environmental conditions," *IEEE Transactions on Sustainable Energy*, vol. 9, pp. 1487–1496, July 2018.

[10] K. R. Bharath and E. Suresh, "Design and implementation of improved fractional open circuit voltage based maximum power point tracking algorithm for photovoltaic applications," *International Journal of Renewable Energy Research-IJRER*, vol. 7, no. 3, 2017.

[11] T. K. Soon and S. Mekhilef, "A fast-converging mppt technique for photovoltaic system under fast-varying solar irradiation and load resistance," *IEEE Transactions on Industrial Informatics*, vol. 11, pp. 176–186, Feb. 2015.

[12] J. Ahmed and Z. Salam, "A modified p&o maximum power point tracking method with reduced steady-state oscillation and improved tracking efficiency," *IEEE Transactions on Sustainable Energy*, vol. 7, pp. 1506–1515, Oct. 2016.

[13] H. A. Sher, A. F. Murtaza, A. Noman, K. E. Addoweesh, K. Al-Haddad, and M. Chiaberge, "A new sensorless hybrid mppt algorithm based on fractional short-circuit current measurement and p&o mppt," *IEEE Transactions on Sustainable Energy*, vol. 6, pp. 1426–1434, Oct. 2015.

[14] M. Taguchi, A. Yano, S. Tohoda, K. Matsuyama, Y. Nakamura, T. Nishiwaki, K. Fujita, and E. Maruyama, "24.7 x0025; record efficiency hit solar cell on thin silicon wafer," *IEEE Journal of Photovoltaics*, vol. 4, pp. 96–99, Jan. 2014.

[15] N. Balakrishnan, T. S. Angel, and R. C. Lekshmi, "Pv-battery system for smart green building using transformer coupled dc-dc converter," in *2016 IEEE 1st International Conference on Power Electronics, Intelligent Control and Energy Systems (ICPEICES)*, pp. 1–6, July 2016.

[16] X. Li, H. Wen, L. Jiang, W. Xiao, Y. Du, and C. Zhao, "An improved mppt method for pv system with fast-converging speed and zero oscillation," *IEEE Transactions on Industry Applications*, vol. 52, pp. 5051–5064, Nov. 2016.

[17] A. Chikh and A. Chandra, "An optimal maximum power point tracking algorithm for pv systems with climatic parameters estimation," *IEEE Transactions on Sustainable Energy*, vol. 6, pp. 644–652, April 2015.

[18] K. R. Bharath, H. Choutapalli, and P. Kanakasabapathy, "Control of bidirectional dc-dc converter in renewable based dc microgrid with improved voltage stability," *International Journal of Renewable Energy Research*, vol. 8, pp. 871–877, 01 2018.

[19] E. Dallago, A. Liberale, D. Miotti, and G. Venchi, "Direct mppt algorithm for pv sources with only voltage measurements," *IEEE Transactions on Power Electronics*, vol. 30, pp. 6742–6750, Dec. 2015.

[20] K. R. Bharath and P. Kanakasabapathy, "Implementation of enhanced perturb and observe maximum power point tracking algorithm to overcome partial shading losses," in *2016 International Conference on Energy Efficient Technologies for Sustainability (ICEETS)*, pp. 62–67, April 2016.

Authorized licensed use limited to: Universite de Technologie de Compiegne (UTC). Downloaded on April 08,2024 at 16:43:14 UTC from IEEE Xplore. Restrictions apply.